

gurps-ink

Typeset unofficial *GURPS* material
User guide and reference, version 0.1.0

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1 What gurps-ink does

gurps-ink is a Typst package for people who write unofficial **GURPS** material: a sourcebook, an adventure or a handout. It does three things.

Task	What gurps-ink gives you	Chapter
Legal notices	The disclaimer and the notice that the Steve Jackson Games online policy asks for, and GURPS in bold italics.	Section 3
Dice and books	Dice such as 2d-1, damage from ST, and references such as GURPS Magic p. 14.	Section 4
Characters	Stat blocks that calculate their point costs.	Section 5

It does not do these things:

- It does not copy the look of **GURPS** books. The Steve Jackson Games online policy does not let you do that (Section 3.4). gurps-ink follows the typographic rules of Steve Jackson Games, and you choose the fonts and layout.
- It does not give legal advice.
- It does not calculate the cost of advantages and disadvantages. You give these costs.
- It does not import characters from GURPS Character Sheet (GCS).

1.1 How to use this manual

Start with Section 2. Then go to the chapter for your task. Each chapter starts with a list of the questions that it answers. Each section answers one question.

In each example, the code is on the left and the result is on the right. Each example in chapters 2 to 5 is complete: copy it into a new file and it compiles. A reference example (Section 7) needs the import line first.

The examples use one story: a race between the Tortoise and the Hare. Section 6 shows the complete one-shot adventure that they come from.

1.2 Find your question

I want to ...	Go to
install gurps-ink	Section 2
add the notices that Steve Jackson Games asks for	Section 3.1
know if my book can look like a GURPS book	Section 3.4
write dice or damage	Section 4.1, Section 4.2
refer to a GURPS book or page	Section 4.3
make a stat block	Section 5.1
know which costs I must give	Section 5.2
hide point costs in a handout	Section 5.3
write a character's numbers in my text	Section 5.4
use one character in many chapters	Section 5.5
keep a character in a point budget	Section 5.6
keep a stat block in one column	Section 5.7
change the look of a stat block	Section 5.8

make a cast list	Section 5.9
understand an error	Section 5.10
see a complete adventure	Section 6
look up a function	Section 7

2 Get started

This chapter answers:

- How do I install gurps-ink?
- What does a first page look like?

2.1 Install gurps-ink

gurps-ink is not on Typst Universe yet. Until it is, install it on your computer:

1. Install [Typst](#) and [just](#).
2. Clone the [repository](#).
3. In the repository, run `just install`.
4. In your document, write `#import "@local/gurps-ink:0.1.0": *`.

When gurps-ink is on Typst Universe, you do not install it. Write `#import "@preview/gurps-ink:0.1.0": *` and Typst downloads it. This also works in the [Typst web app](#). The examples in this manual use this import.

2.2 Make your first page

Copy this example into a new file and compile it. It makes a character, shows its stat block and adds the disclaimer.

```
#import "@preview/gurps-ink:0.1.0": *

#let tortoise = character(
  name: "The Tortoise",
  ht: 13,
  basic-move: 1,
  skill("Running", 14, "HT/A"),
)

#stat-block(tortoise)

The Tortoise runs at Move
#level-of(tortoise, "Basic Move").

#text(size: 7pt, sjgames-disclaimer)
```

The Tortoise (14 points)

ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 13 [30].
HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 13 [0].
Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8.
SM 0; DR 0; thr 1d-2; sw 1d.
Skills: Running-14 [4].

The Tortoise runs at Move 1.

The material presented here is my original creation, intended for use with the [GURPS](#) system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

The import line gives you all functions of gurps-ink. The next three chapters show what you can do with them.

3 Legal notices and trademarks

This chapter answers:

- Which notices must my material have? (Section 3.1)
- What if my material is a free game aid? (Section 3.2)
- How do I write “GURPS”? (Section 3.3)
- Can my book look like a GURPS book? (Section 3.4)
- Where does the text of the notices come from? (Section 3.5)

gurps-ink gives you text. It does not give legal advice. Read the [Steve Jackson Games online policy](#) before you publish. The policy is for free material. To sell material, read its section VII, “Licensed Use”.

3.1 Add the disclaimer and the notice

Section IV of the online policy, “Notices and Disclaimers”, asks for two texts on material that is not official:

- A *disclaimer*: the material is not official and Steve Jackson Games does not endorse it. Use `sjgames-disclaimer`.
- A *notice*: **GURPS** is a trademark of Steve Jackson Games. Use `sjgames-notice`.

Both texts have links to the Steve Jackson Games pages, as the policy asks. Put them on the title page or on the last page. You can make them small.

```
#import "@preview/gurps-ink:0.1.0": *  
  
#set text(size: 7pt)  
#sjgames-disclaimer  
  
#sjgames-notice
```

The material presented here is my original creation, intended for use with the **GURPS** system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

GURPS is a registered trademark of [Steve Jackson Games](#), and is copyrighted by Steve Jackson Games. All rights are reserved by Steve Jackson Games. This material is used here in accordance with the [SJ Games online policy](#).

3.2 Add a game aid notice

`sjgames-game-aid` is a notice for a free game aid, such as a character sheet or a tool. Give it the name of the author. The online policy does not give this text. Many free **GURPS** game aids use it, for example the GURPS system for Foundry VTT.

```
#import "@preview/gurps-ink:0.1.0": *  
  
#set text(size: 7pt)  
#sjgames-game-aid[Jane Doe]
```

GURPS is a trademark of [Steve Jackson Games](#), and its rules and art are copyrighted by [Steve Jackson Games](#). All rights are reserved by Steve Jackson Games. This game aid is the original creation of Jane Doe and is released for free distribution, and not for resale, under the permissions granted in the [Steve Jackson Games Online Policy](#).

3.3 Write GURPS and other trademarks

The online policy asks you to show each trademark in bold, italic or colour each time you use it. `gurps` writes **GURPS** in bold italics. `sjgames` writes Steve Jackson Games. For other trademarks, use `strong` and `emph`.

```
#import "@preview/gurps-ink:0.1.0": *  
  
This is a one-shot for #gurps.  
#sjgames publishes  
#strong(emph[Munchkin]) too.
```

This is a one-shot for **GURPS**. Steve Jackson Games publishes **Munchkin** too.

Do not put #gurps in italic text. In italic text, emph makes it upright.

3.4 Use your own look

Section II of the online policy says that if your material looks like a Steve Jackson Games product, it is “over the line”. For example, do not copy the covers of **GURPS** books.

gurps-ink does not copy the look of **GURPS** books. It sets no fonts, colours or page layout. It follows the typographic rules of Steve Jackson Games: how to write dice, book titles, page references and stat blocks. Choose your own fonts, colours and layout.

3.5 Where the text comes from

Function	Source
sjgames-disclaimer	The example disclaimer in section IV of the online policy .
sjgames-notice	The example notice in section IV of the online policy , without the sentence about art.
sjgames-game-aid	Not from the policy. The notice that many free GURPS game aids use.
gurps, sjgames	The trademark rules in the online policy .

4 Dice and book references

This chapter answers:

- How do I write dice, such as 3d or 2d-1? (Section 4.1)
- How do I write the damage for a ST? (Section 4.2)
- How do I refer to a **GURPS** book or a page of it? (Section 4.3)
- Where do these rules come from? (Section 4.4)

4.1 Write dice

Use `dice(count, modifier)`. The modifier is optional. A negative modifier has a true minus sign, not a hyphen. The result does not break across lines.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
The Hare rolls #dice(3) against  
its self-control number.  
Its kick does #dice(1, -4) cr.  
With a symbol: #dice(2, $n$).
```

The Hare rolls 3d against its self-control number. Its kick does 1d-4 cr. With a symbol: 2d+n.

4.2 Write damage from ST

`thrust(st)` and `swing(st)` give the basic damage for a ST, from the table on p. B16. They cover ST 1 to 100. Spread the result into dice with ...

```
#import "@preview/gurps-ink:0.1.0": *
```

```
At ST 6, the Hare kicks for  
#dice(..thrust(6)) cr.  
At ST 13, a sword swings for  
#dice(..swing(13)) cut.
```

At ST 6, the Hare kicks for 1d-4 cr. At ST 13, a sword swings for 2d-1 cut.

A character from `character()` has its damage in `thr` and `sw` (Section 5.4).

4.3 Refer to a GURPS book

Use `gurps-book(title, pages)`. Do not write “GURPS” in the title: the function adds it. The pages are optional. One page gets “p.”; more pages get “pp.”.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
You need the #gurps-book("Basic Set").  
See #gurps-book("Magic", 14) and  
#gurps-book("Fantasy", (12, 20)).
```

You need the **GURPS Basic Set**. See **GURPS Magic** p. 14 and **GURPS Fantasy** pp. 12, 20.

For a page of the *Basic Set*, the style of Steve Jackson Games is “p. B348”. Write it as text: p.~B348. The ~ keeps “p.” and the number on one line.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
Roll a Quick Contest (p.~B348).
```

Roll a Quick Contest (p. B348).

4.4 Where the rules come from

The dice notation and the bold italic book titles follow the [Steve Jackson Games Authors’ Guidelines: Editorial Style](#). The damage table is on p. B16 of the **GURPS Basic Set**.

5 Characters and stat blocks

This chapter answers:

- How do I make a character and show its stat block? (Section 5.1)
- Which point costs does gurps-ink calculate? (Section 5.2)
- How do I hide the point costs? (Section 5.3)
- How do I write a character's numbers in my text? (Section 5.4)
- How do I use one character in many chapters? (Section 5.5)
- How do I keep a character in a point budget? (Section 5.6)
- How do I stop a column break from splitting a stat block? (Section 5.7)
- How do I change the look of a stat block? (Section 5.8)
- How do I make a cast list or another layout? (Section 5.9)
- What does an error message mean? (Section 5.10)

A character has two steps. `character()` makes a dictionary and calculates the costs. `stat-block()` shows the dictionary. Because the two steps are separate, you can make a character once and use it in many places.

5.1 Make a character and show it

Give the attributes as named arguments. Give only the attributes that are not the default. Give the traits, skills, spells and attacks as positional arguments, in any order.

```
#import "@preview/gurps-ink:0.1.0": *

#let hare = character(
  name: "The Hare",
  st: 6, dx: 13, ht: 11,
  sm: -2,
  basic-move: 9,
  disadvantage(
    "Overconfidence (12)",
    points: -5,
  ),
  quirk("Naps after lunch"),
  skill("Running", 13, "HT/A"),
  melee-attack("Kick", 13,
    [#dice(1, -4) cr], reach: "C"),
)
#stat-block(hare)
```

The Hare (47 points)

ST 6 [-40]; DX 13 [60]; IQ 10 [0]; HT 11 [10].
HP 6 [0]; Will 10 [0]; Per 10 [0]; FP 11 [0].
Basic Speed 6.00 [0]; Basic Move 9 [15]; Dodge 9.
SM -2; DR 0; thr 1d-4; sw 1d-3.

Disadvantages: Overconfidence (12) [-5].

Quirks: Naps after lunch [-1].

Skills: Running-13 [8].

Attacks:

Kick (13): 1d-4 cr. Reach C.

The functions for traits are advantage, disadvantage, perk and quirk. For skills and spells, they are `skill` and `spell`. For attacks, they are `melee-attack` and `ranged-attack`. The reference (Section 7.3) shows all their arguments.

5.2 Know which costs are calculated

<code>character()</code> calculates	You give
The cost of each attribute and secondary characteristic	The cost of each advantage and disadvantage, with points:
The cost of a skill or spell, if you give its base and difficulty, such as "HT/A"	The cost of a skill or spell, if you give a number instead
Dodge, from Basic Speed	The skill level and damage of each attack
Thrust and swing damage, from ST 1 to 100	Damage above ST 100, with <code>thr:</code> and <code>sw:</code>

The point total	Discounts, such as the SM discount to ST
-----------------	--

A value that you give always replaces the calculated value. To give the cost of an attribute, give a dictionary: `st: (level: 20, points: 90)`. If an advantage or disadvantage has no points:, the stat block shows [?] and the total counts it as 0.

```
#import "@preview/gurps-ink:0.1.0": *

#stat-block(character(
  name: "The Tortoise",
  ht: 13,
  advantage("Fit", points: 5),
  advantage("Patience"),
  skill("Running", 14, "HT/A"),
  skill("Fast-Talk", 11, 2),
))
```

The Tortoise (41 points)
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 13 [30].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 13 [0].
 Basic Speed 5.75 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Advantages: Fit [5]; Patience [?].
Skills: Fast-Talk-11 [2]; Running-14 [4].

A skill can also have another skill of the same character as its base, such as "Running/E".

5.3 Hide point costs

Players do not need point costs in a handout or a bestiary. Use `show-points: false`.

```
#import "@preview/gurps-ink:0.1.0": *

#stat-block(
  character(
    name: "The Hare",
    dx: 13,
    skill("Running", 13, "HT/A"),
  ),
  show-points: false,
)
```

The Hare
 ST 10; DX 13; IQ 10; HT 10.
 HP 10; Will 10; Per 10; FP 10.
 Basic Speed 5.75; Basic Move 5; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Skills: Running-13.

5.4 Write a character's numbers in your text

Use `level-of(char, name)` for the level of an attribute, skill or spell. Use `char.thr` and `char.sw` for the damage. If you change the character, the text changes too.

```
#import "@preview/gurps-ink:0.1.0": *

#let hare = character(
  st: 6, dx: 13,
  basic-move: 9,
  skill("Running", 13, "HT/A"),
)

The Hare has Move
#level-of(hare, "Basic Move") and
Running-#level-of(hare, "Running").
It kicks for #dice(..hare.thr) cr.
```

The Hare has Move 9 and Running-13. It kicks for 1d-4 cr.

5.5 Use one character in many chapters

Put all characters in one dictionary. Then each chapter can use the same characters.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
#let npcs = (
  tortoise: character(
    name: "The Tortoise", ht: 13),
  hare: character(
    name: "The Hare", dx: 13),
)
```

The Tortoise has
`#level-of(npcs.tortoise, "HT")` HT.
 The Hare has
`#level-of(npcs.hare, "DX")` DX.

The Tortoise has 13 HT. The Hare has 13 DX.

Put the import line and the `#let npcs = (..)` lines in a file, for example `npcs.typ`. In each chapter, write `#import "npcs.typ": npcs`.

5.6 Keep a character in a point budget

Use `assert` with `total-points`. If the character costs too much, compilation stops and shows your message. This is useful for a pregenerated character or a template.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
#let tortoise = character(
  name: "The Tortoise",
  iq: 11, ht: 13,
  basic-move: 1,
)
#assert(
  total-points(tortoise) <= 50,
  message: "Over 50 points",
)
#stat-block(tortoise)
```

The Tortoise (30 points)
 ST 10 [0]; DX 10 [0]; IQ 11 [20]; HT 13 [30].
 HP 10 [0]; Will 11 [0]; Per 11 [0]; FP 13 [0].
 Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.

5.7 Keep a stat block in one column

A stat block can break across columns and pages. To prevent this, put it in a block that cannot break.

```
#import "@preview/gurps-ink:0.1.0": *
```

```
#block(
  breakable: false,
  stat-block(character(
    name: "The Hare")),
)
```

The Hare (0 points)
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].
 Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.

5.8 Change the look of a stat block

Use the hooks of `stat-block()`. The title hook makes the first line. The section hook makes each list, such as “Skills”. To frame a stat block, put it in a block with a `stroke`.

```
#import "@preview/gurps-ink:0.1.0": *

#block(stroke: 0.5pt, inset: 6pt,
  stat-block(
    character(
      name: "The Hare",
      skill("Running", 13, "HT/A"),
    ),
    title: (char, total) => smallcaps(
      [#char.name, #total points],
    ),
    section: (label, body) => [
      #emph(label) --- #body
    ],
  ),
))
```

THE HARE, 12 POINTS
 ST 10 [0]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
 HP 10 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].
 Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
 SM 0; DR 0; thr 1d-2; sw 1d.
Skills — Running-13 [12].

5.9 Make a cast list or another layout

Write a function that takes the dictionary from `character()`. The reference for `character()` (Section 7.3) shows all keys of this dictionary.

```
#import "@preview/gurps-ink:0.1.0": *

#let cast(char) = {
  let skills = char.skills
  .map(s => [#s.name\-#s.level])
  [*#char.name* (#total-points(char)
    points): #skills.join[, ].]
}

#cast(character(
  name: "The Tortoise", ht: 13,
  skill("Running", 14, "HT/A"),
))

#cast(character(
  name: "The Hare", dx: 13,
  skill("Running", 13, "HT/A"),
))
```

The Tortoise (34 points): Running-14.

The Hare (72 points): Running-13.

When Typst gets user-defined elements, `stat-block` will become one. Then show and set rules can change it, as they change built-in elements.

5.10 Understand an error

When an input is wrong, compilation stops. The message tells you what is wrong. This table shows the frequent messages.

The message contains	Do this
is too low to buy	Make the skill level higher, or give the points as a number.
cannot work out the cost of	Make sure that the base ("HT" in "HT/A") is the name of an attribute, skill or spell of the same character.
more than one skill or spell is called	Give each skill and spell a different name.
got unknown argument(s)	Make sure that you spell the argument correctly.
covers ST 1 to 100	Give <code>thr:</code> and <code>sw:</code> to <code>character()</code> .

points must be zero or negative	Give a disadvantage a negative cost, for example points : -10.
---------------------------------	---

6 Example: The Tortoise and the Hare

This one-shot adventure uses each part of `gurps-ink`. Its source is [docs/example/tortoise-and-hare.typ](#). The PDF is [docs/tortoise-and-hare.pdf](#).

The Tortoise and the Hare A one-shot adventure for <i>GURPS</i>. One player is the Tortoise. The GM is the Hare. It takes about 30 minutes. You need the <i>GURPS Basic Set</i>. The Challenge The Hare boasts that no animal in the meadow can beat it in a race. The Tortoise accepts the challenge. The race starts at the old oak and ends at the mill pond, one mile away. The Characters The player uses the Tortoise. It is built on 50 points. The Tortoise (50 points) ST 10 [0]; DX 10 [0]; IQ 11 [20]; HT 13 [30]. HP 10 [0]; Will 11 [0]; Per 11 [0]; FP 13 [0]. Basic Speed 5.75 [0]; Basic Move 1 [-20]; Dodge 8. SM 0; DR 3 (shell); thr 1d-2; sw 1d. Advantages: Damage Resistance (Tough Shell) 3 [15]; Fit [5]. Disadvantages: Stubbornness [-5]. Quirks: Never hurries [-1]. Skills: Fast-Talk-11 [2]; Running-14 [4]. The GM plays the Hare. The Hare is fast, but it has Overconfidence (p. B148). The Hare (47 points) ST 6 [-40]; DX 13 [60]; IQ 10 [0]; HT 11 [10]. HP 6 [0]; Will 10 [0]; Per 10 [0]; FP 11 [0]. Basic Speed 6.00 [0]; Basic Move 9 [15]; Dodge 9. SM -2; DR 0; thr 1d-4; sw 1d-3. Disadvantages: Overconfidence (12) [-5]. Quirks: Naps after lunch [-1]. Skills: Running-13 [8]. Attacks: Kick (13): 1d-4 cr. Reach C. The Race The race has three steps. 1. Taunt (optional). Before the start, the Tortoise can mock the Hare. Roll a Quick Contest (p. B348) of the Tortoise's Fast-Talk-11 against the Hare's Will of 10. If the Tortoise wins, the Hare takes -2 on its roll in step 2. 2. The nap. Halfway, the Hare is far ahead. The GM rolls 3d against the Hare's self-control number of 12 (p. B120). On a failure, the Hare stops for a nap.	3. The finish. Roll a Quick Contest of Running (p. B218): the Tortoise's Running-14 against the Hare's Running-13. The Hare gets +4 for its speed. If it took a nap, it gets -6 instead. The winner reaches the pond first. On a tie, the Tortoise wins by the length of its nose. After the Race If the Tortoise wins, the meadow cheers. Give the player 1 character point. If the Hare wins, it boasts for a week. The player still gets 1 character point: slow and steady learns a lesson too.
1	2

In the adventure	How it is made
Fonts, colours and page size of its own	Section 3.4
<i>GURPS</i> and <i>GURPS Basic Set</i> in the introduction	Section 3.3, Section 4.3
Two stat blocks that do not break	Section 5.1, Section 5.7
The Tortoise is built on 50 points	Section 5.6
Skill levels in the text of the race	Section 5.4
3d for the nap roll	Section 4.1
The disclaimer and the notice at the end	Section 3.1

The source follows.

```
#import "@preview/gurps-ink:0.1.0": *

// Your own look: gurps-ink sets no fonts, colours or page layout.
#set document(title: "The Tortoise and the Hare")
#set page(paper: "a5", margin: (x: 15mm, y: 18mm), numbering: "1")
#set text(size: 9.5pt)
#set par(justify: true)
#show heading: set text(fill: rgb("#2f5d50"))
#show heading.where(level: 1): set text(size: 16pt)

// The two characters. Each section below uses them again.
#let tortoise = character(
  name: "The Tortoise",
  iq: 11,
```

```

    ht: 13,
    basic-move: 1,
    dr: [3 (shell)],
    advantage("Damage Resistance (Tough Shell)", points: 15, level: 3),
    advantage("Fit", points: 5),
    disadvantage("Stubbornness", points: -5),
    quirk("Never hurries"),
    skill("Running", 14, "HT/A"),
    skill("Fast-Talk", 11, "IQ/A"),
  )
#let hare = character(
  name: "The Hare",
  st: 6,
  dx: 13,
  ht: 11,
  sm: -2,
  basic-move: 9,
  disadvantage("Overconfidence (12)", points: -5),
  quirk("Naps after lunch"),
  skill("Running", 13, "HT/A"),
  melee-attack("Kick", 13, [#dice(..thrust(6)) cr], reach: "C"),
)

// The player builds the Tortoise on 50 points. Compilation stops if a
// change makes it cost more.
#assert(total-points(tortoise) <= 50, message: "The Tortoise is over 50 points")

```

== The Tortoise and the Hare

A one-shot adventure for #gurps. One player is the Tortoise. The GM is the Hare. It takes about 30 minutes. You need the #gurps-book("Basic Set").

== The Challenge

The Hare boasts that no animal in the meadow can beat it in a race. The Tortoise accepts the challenge. The race starts at the old oak and ends at the mill pond, one mile away.

== The Characters

The player uses the Tortoise. It is built on 50 points.

```
#block(breakable: false, stat-block(tortoise))
```

The GM plays the Hare. The Hare is fast, but it has Overconfidence (p. B148).

```
#block(breakable: false, stat-block(hare))
```

== The Race

The race has three steps.

- + ***Taunt (optional)*** Before the start, the Tortoise can mock the Hare. Roll a Quick Contest (p. B348) of the Tortoise's Fast-Talk-#level-of(tortoise, "Fast-Talk") against the Hare's Will of #level-of(hare, "Will"). If the Tortoise wins, the Hare takes -2 on its roll in step 2.
- + ***The nap.*** Halfway, the Hare is far ahead. The GM rolls #dice(3) against the Hare's self-control number of 12 (p. B120). On a failure, the Hare stops for a nap.
- + ***The finish.*** Roll a Quick Contest of Running (p. B218): the Tortoise's Running-#level-of(tortoise, "Running") against the Hare's Running-#level-of(hare, "Running"). The Hare gets +4 for its speed. If it took a nap, it gets -6 instead. The winner reaches the pond first. On a tie, the Tortoise wins by the length of its nose.

== After the Race

If the Tortoise wins, the meadow cheers. Give the player 1 character point. If the Hare wins, it boasts for a week. The player still gets 1 character point: slow and steady learns a lesson too.

```

#v(1fr)
#line(length: 100%, stroke: 0.4pt)

```

```
#set text(size: 7pt)
#sjgames-disclaimer
#sjgames-notice
```

7 Reference

This chapter lists each function, in the same groups as chapters 3 to 5. Each example needs the import line first.

7.1 Legal notices and trademarks

7.1.1 sjgames-game-aid

A notice for a free game aid that has no official license. The SJ Games online policy does not give this text. Many free GURPS game aids use it.

```
#sjgames-game-aid[Jane Doe]
```

GURPS is a trademark of [Steve Jackson Games](#), and its rules and art are copyrighted by [Steve Jackson Games](#). All rights are reserved by Steve Jackson Games. This game aid is the original creation of Jane Doe and is released for free distribution, and not for resale, under the permissions granted in the [Steve Jackson Games Online Policy](#).

Parameters

```
sjgames-game-aid(author: str content) -> content
```

author `str` or `content`

The author of the game aid.

7.1.2 sjgames-disclaimer `content`

The disclaimer from the SJ Games online policy. It says that the material is yours and not official.

```
#sjgames-disclaimer
```

The material presented here is my original creation, intended for use with the **GURPS** system from [Steve Jackson Games](#). This material is not official and is not endorsed by Steve Jackson Games.

7.1.3 sjgames-notice content

The trademark notice from the SJ Games online policy. It does not mention art, because the package includes no SJ Games art.

#sjgames-notice

GURPS is a registered trademark of [Steve Jackson Games](#), and is copyrighted by Steve Jackson Games. All rights are reserved by Steve Jackson Games. This material is used here in accordance with the SJ Games [online policy](#).

7.1.4 gurps content

The name **GURPS** in bold italics. The SJ Games online policy asks for this style. Use it as you use other content.

#gurps is a generic, universal roleplaying system.

GURPS is a generic, universal roleplaying system.

To link it to the GURPS home page, put it in link: #link("http://www.sjgames.com/gurps/", gurps).

7.1.5 sjgames content

The name "Steve Jackson Games".

#gurps is published by #sjgames.

GURPS is published by Steve Jackson Games.

7.2 Dice and book references

7.2.1 dice

Dice in the notation of the GURPS books. The result does not break across lines.

```
#dice(3) \
#dice(2, 1) \
#dice(4, -1) \
#dice(1, 0)
```

3d
2d+1
4d-1
1d

The modifier is optional. A zero modifier shows nothing. A negative modifier has a true minus sign. A content modifier, such as \$m\$, gets a +.

To write basic damage, spread `thrust()` or `swing()` into it: `#dice(..thrust(13))`.

Parameters

```
dice(  
  count: int content,  
  ..modifier: int content  
) -> content
```

count int or content

The number of dice: an integer of 0 or more, or content such as \$n\$.

..modifier int or content

The modifier. It is optional; the default is 0.

7.2.2 thrust

The thrust damage for a ST, as a (count, modifier) pair. Spread it into `dice()`.

```
#thrust(10) \  
#dice(..thrust(10))
```

```
(1, -2)  
1d-2
```

It covers ST 1 to 100. Above ST 40, the Basic Set table has a row for each 5 ST; between two rows, the lower row applies. For a ST that is not in this range, compilation stops: give thr: to `character()`.

Parameters

```
thrust(st: int) -> array
```

st int

The striking strength.

7.2.3 swing

The swing damage for a ST, as a (count, modifier) pair. Spread it into `dice()`. It covers the same ST as `thrust()`.

```
#dice(..swing(10))
```

```
1d
```

Parameters

```
swing(st: int) -> array
```

st **int**

The striking strength.

7.2.4 gurps-book

A reference to a GURPS book in the house style of SJ Games: the title in bold italics, then the pages if you give them.

```
#gurps-book("High Tech") \
#gurps-book("Zombies", 3) \
#gurps-book("Warehouse 23", "1, 3-5") \
#gurps-book("Basic Set", (16, 170))
```

GURPS High Tech
GURPS Zombies p. 3
GURPS Warehouse 23 pp. 1, 3-5
GURPS Basic Set pp. 16, 170

One page (an int, or a string of digits) gets “p.”. All other pages get “pp.”. Commas join the pages in an array.

Parameters

```
gurps-book(
  title: str content,
  ..pages: int str content array
) -> content
```

title **str** or **content**

The title of the book, without “GURPS”.

..pages **int** or **str** or **content** or **array**

The page or pages. They are optional.

7.3 Characters and stat blocks

7.3.1 character

Makes a character. Give only the values that are not the default.

Give each attribute as a level. `character()` calculates its cost from the Basic Set rules. To give a different cost, such as an SM discount to ST, give a dictionary: `st: (level: 20, points: 90)`.

Give the traits, skills, spells and attacks as positional arguments.

```
#let napoleon = character(
  name: "Napoleon",
  st: 9, hp: 12,
  advantage("Natural afro", points: 1),
  quirk("Big teeth"),
  skill("Nunchuck", 16, "DX/E"),
)
#napoleon.attributes.HP \
#total-points(napoleon) points
```

(level: 12, points: 6)
16 points

The result is a dictionary. Its keys are public API. Read them, or use `level-of()`, to quote numbers or to make your own layout. The keys are:

- kind: "character"; name.
- attributes: keyed ST, DX, IQ, HT, HP, Will, Per, FP, Basic Speed, Basic Move, in that order; each (level:, points:).
- dodge, sm, dr, thr, sw: as given or calculated.
- advantages, perks, disadvantages, quirks: arrays of (kind:, name:, points:, level:) in the order given.
- skills, spells: arrays of (kind:, name:, level:, points:, base:, difficulty:). If the skill has a base, points is calculated.
- attacks: arrays of (kind:, name:, level:, damage:, notes:) plus reach (melee) or range (ranged).

A value that is not known is none.

Parameters

```
character(  
  name: str content none ,  
  st: int dictionary ,  
  dx: int dictionary ,  
  iq: int dictionary ,  
  ht: int dictionary ,  
  hp: auto int dictionary ,  
  will: auto int dictionary ,  
  per: auto int dictionary ,  
  fp: auto int dictionary ,  
  basic-speed: auto int float dictionary ,  
  basic-move: auto int dictionary ,  
  dodge: auto int ,  
  sm: int ,  
  dr: int str content ,  
  thr: auto array content ,  
  sw: auto array content ,  
  ..traits: dictionary  
) -> dictionary
```

name str or content or none

The name. The stat block shows it in the title.

Default: none

basic-speed auto or int or float or dictionary

A multiple of 0.25.

Default: auto

dodge auto or int

The default is [Basic Speed] + 3. It has no point cost.

Default: auto

sm `int`

The Size Modifier.

Default: `0`

dr `int` or `str` or `content`

The Damage Resistance, such as 2 or [3 (torso only)].

Default: `0`

thr `auto` or `array` or `content`

The thrust damage: a (count, modifier) pair or content. The default is `thrust()` of ST.

Default: `auto`

sw `auto` or `array` or `content`

The swing damage, as for thr. The default is `swing()` of ST.

Default: `auto`

..traits `dictionary`

Results of `advantage()`, `disadvantage()`, `perk()`, `quirk()`, `skill()`, `spell()`, `melee-attack()` and `ranged-attack()`, in any order.

7.3.2 stat-block

Typesets a character from `character()` as a stat block, in the style of the SJ Games books.

The lines are: the title (name and point total), ST/DX/IQ/HT, HP/Will/Per/FP, Basic Speed/Basic Move/Dodge and SM/DR/thr/sw. Then come the lists: advantages, perks, disadvantages, quirks, skills, spells and attacks. Each list is in alphabetical order. An empty list does not show.

```
#stat-block(character(
  name: "Napoleon",
  st: 9, hp: 12,
  advantage("Natural afro", points: 1),
  quirk("Big teeth"),
  skill("Nunchuck", 16, "DX/E"),
  melee-attack("Punch", 18, [#dice(5) cr],
reach: "C, 1",
  notes: [Believe it!]),
))
```

Napoleon (16 points)
ST 9 [-10]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
HP 12 [6]; Will 10 [0]; Per 10 [0]; FP 10 [0].
Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
SM 0; DR 0; thr 1d-2; sw 1d-1.
Advantages: Natural afro [1].
Quirks: Big teeth [-1].
Skills: Nunchuck-16 [20].
Attacks:
Punch (18): 5d cr. Reach C, 1. Believe it!

To frame it, put it in a block: `#block(stroke: 0.5pt, inset: 8pt)[#stat-block(c)]`.

Two hooks change parts of the stat block:

```
#stat-block(
  character(name: "Guard", st: 11,
skill("Spear", 12, "DX/A")),
  title: (char, total) =>
smallcaps[#char.name, #total points],
  section: (label, body) => [#emph(label)
--- #body],
)
```

GUARD, 18 POINTS
ST 11 [10]; DX 10 [0]; IQ 10 [0]; HT 10 [0].
HP 11 [0]; Will 10 [0]; Per 10 [0]; FP 10 [0].
Basic Speed 5.00 [0]; Basic Move 5 [0]; Dodge 8.
SM 0; DR 0; thr 1d-1; sw 1d+1.
Skills — Spear-12 [8].

For a different layout, write your own function that uses the dictionary from `character()`. The manual shows how.

Parameters

```
stat-block(
  char: dictionary,
  show-points: bool,
  title: auto none function,
  section: auto function
) -> content
```

char dictionary

A result of `character()`.

show-points `bool`

Show the point costs ([10]) and the total. Use `false` for a handout or a bestiary.

Default: `true`

title `auto` or `none` or `function`

Makes the first line. The call is `title(char, total)`. `total` is the point total, or `none` if `show-points` is `false`. Return content or a string. To remove the line, return `none` or give `title: none`. `auto` is this function:

```
(char, total) => {  
  let points = if total != none [#total points]  
  if char.name == none { points }  
  else [#strong(char.name)#if points != none [ (#points)]]  
}
```

Default: `auto`

section `auto` or `function`

Makes each list: Advantages, Perks, Disadvantages, Quirks, Skills, Spells and Attacks. The call is `section(label, body)`. `body` is the finished list: the entries, with semicolons between them and a full stop at the end. For attacks, `body` is a block with one paragraph for each attack. Return content or a string. To remove the list, return `none`. `auto` is `(label, body) => [#strong[#label:] #body]`.

Default: `auto`

7.3.3 advantage

An advantage.

```
#let afro = advantage("Natural afro",  
points: 1)  
#let magery = advantage("Magery", points:  
25, level: 2)  
#afro.points, #magery.level
```

1, 2

Parameters

```
advantage(  
  name: str content,  
  points: int none,  
  level: int none  
) -> dictionary
```

points `int` or `none`

The point cost. If it is `none`, the stat block shows [?] and the total counts it as 0.

Default: `none`

level `int` or `none`

The level of an advantage that has levels, such as Magery 2.

Default: `none`

7.3.4 disadvantage

A disadvantage. If you give a point cost, it must be zero or negative.

```
#disadvantage("Bad Temper", points:  
-10).points
```

-10

Parameters

```
disadvantage(  
    name: str content,  
    points: int none,  
    level: int none  
) -> dictionary
```

points `int` or `none`

The point cost, zero or less. If it is none, the stat block shows [?] and the total counts it as 0.

Default: `none`

level `int` or `none`

The level of a disadvantage that has levels.

Default: `none`

7.3.5 perk

A perk. Its cost is always 1 point.

Parameters

```
perk(name: str content) -> dictionary
```

7.3.6 quirk

A quirk. Its cost is always -1 point.

Parameters

```
quirk(name: str content) -> dictionary
```

7.3.7 skill

A skill at a given level.

The third argument is the cost. It is optional. Give it in one of two forms:

- The points, such as 4.
- The base and the difficulty, such as "DX/E" or "Per/Average". Then `character()` calculates the points.

The base can be an attribute, or a skill or spell of the same character ("Typst/H"). The match uses the plain text of skill and spell names, so `[_Typst_]` matches "Typst". If a skill is a base for other skills, give it a name that has plain text.

The difficulty is E, A, H, VH or W, or the full word (Easy, Average, Hard, Very Hard, Wildcard).

```
#skill("Stealth", 12, "DX/A") \  
#skill("Brawling", 14, 4) \  
#skill("Carousing", 10)
```

```
(  
  kind: "skill",  
  name: "Stealth",  
  level: 12,  
  points: none,  
  base: "DX",  
  difficulty: "A",  
)  
(  
  kind: "skill",  
  name: "Brawling",  
  level: 14,  
  points: 4,  
  base: none,  
  difficulty: none,  
)  
(  
  kind: "skill",  
  name: "Carousing",  
  level: 10,  
  points: none,  
  base: none,  
  difficulty: none,  
)
```

Parameters

```
skill(  
  name: str content,  
  level: int,  
  ..cost: int str  
) -> dictionary
```

level int

The skill level.

..cost `int` or `str`

The cost: points, or "<base>/<difficulty>". It is optional.

7.3.8 spell

A spell. It has the same arguments as `skill()`.

```
#spell("Fireball", 14, "IQ/H")
```

```
(  
  kind: "spell",  
  name: "Fireball",  
  level: 14,  
  points: none,  
  base: "IQ",  
  difficulty: "H",  
)
```

Parameters

```
spell(  
  name: str content,  
  level: int,  
  ..cost: int str  
) -> dictionary
```

7.3.9 melee-attack

A melee attack.

```
#melee-attack("Punch", 18, [#dice(1, -1)  
cr], reach: "C")
```

```
(  
  kind: "melee-attack",  
  name: "Punch",  
  level: 18,  
  damage: sequence(box(body: sequence([1,  
[d], [-1]]), [ ], [cr])),  
  notes: none,  
  reach: "C",  
)
```

Parameters

```
melee-attack(  
  name: str content,  
  level: int,  
  damage: str content,  
  reach: str content none,  
  notes: str content none  
) -> dictionary
```

level `int`

The skill level with the attack.

damage `str` or `content`

The damage, such as `[#dice(2, 1) cut]`.

notes `str` or `content` or `none`

Notes. The stat block shows them after the attack.

Default: `none`

7.3.10 ranged-attack

A ranged attack.

```
#ranged-attack("Bow", 14, [#dice(1, 2) imp],  
range: "150/200")
```

```
(  
  kind: "ranged-attack",  
  name: "Bow",  
  level: 14,  
  damage: sequence(box(body: sequence([1,  
[d], [+2]]), [ ], [imp])),  
  notes: none,  
  range: "150/200",  
)
```

Parameters

```
ranged-attack(  
  name: str content ,  
  level: int ,  
  damage: str content ,  
  range: str content none ,  
  notes: str content none  
) -> dictionary
```

7.3.11 level-of

The level of an attribute, skill or spell, found by its name. For a skill or spell, it uses the plain text of the name. It also knows "Dodge" and "SM".

Compilation stops if no name matches, or if two names match. The message lists the names that the character has.

```
#let c = character(dx: 12, skill("Stealth",
13, "DX/A"))
#level-of(c, "Basic Speed"), #level-of(c,
"Stealth")
```

5.5, 13

Parameters

```
level-of(
  char: dictionary,
  name: str
) -> int float
```

char dictionary

A result of `character()`.

name str

The name, such as "ST", "Basic Move" or "Stealth".

7.3.12 total-points

The point total of a character: the sum of all known costs. A trait that has no cost counts as 0.

```
#total-points(character(st: 12, dx: 11))
```

40

Parameters

```
total-points(char: dictionary) -> int
```

char dictionary

A result of `character()`.

8 Legal

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